



A Brief History of Time

The Detective Story of the Universe



Based on the work of Stephen Hawking.

An exploration of the ultimate questions:
Where did the universe come from?
How do we know what we know?
And where is it all going?

Our goal is nothing less than a complete description of the universe we live in.

Turtles All the Way Down



ANCIENT COSMOLOGY:
THE INFINITE REGRESS



MODERN OBSERVATION:
ISOLATED EARTH

A famous scientist once explained the orbits of the earth and sun, only to be told by an old lady:

"What you have told us is rubbish. The world is really a flat plate supported on the back of a giant tortoise."

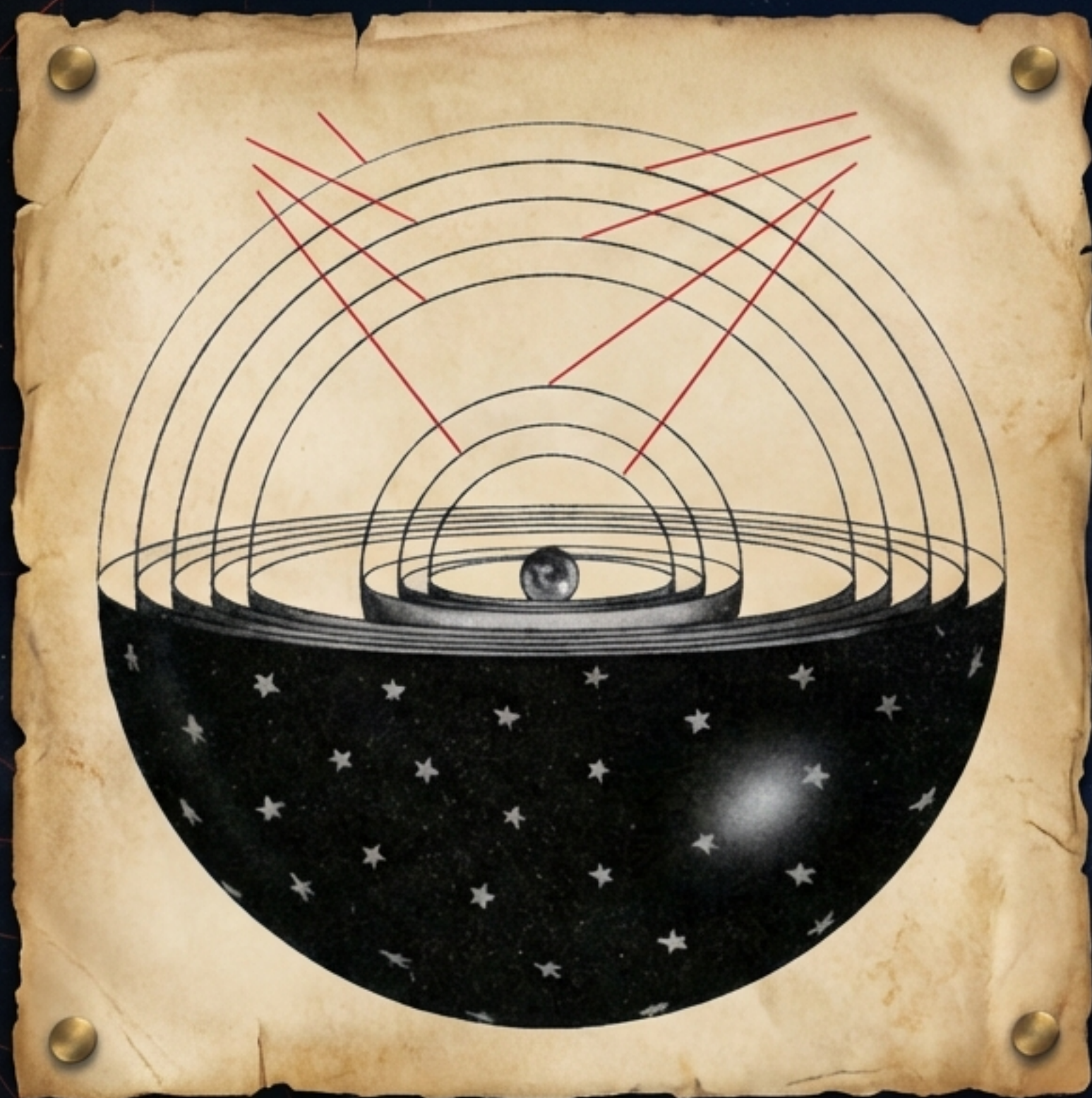
When asked what the tortoise stands on, she replied, "It's turtles all the way down!"

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KEY QUESTION:

Most find this ridiculous, but why do we think we know better? What is the nature of time? Is the universe infinite?

We are peeling back layers of reality.
Yesterday's 'obvious' truth is today's myth.



The Ptolemaic model: A universe wrapped in crystal spheres, safe and centered on us.

The Crystal Spheres

Aristotle (340 BC) argued the Earth was round based on eclipses and the North Star's position, but believed it was stationary.

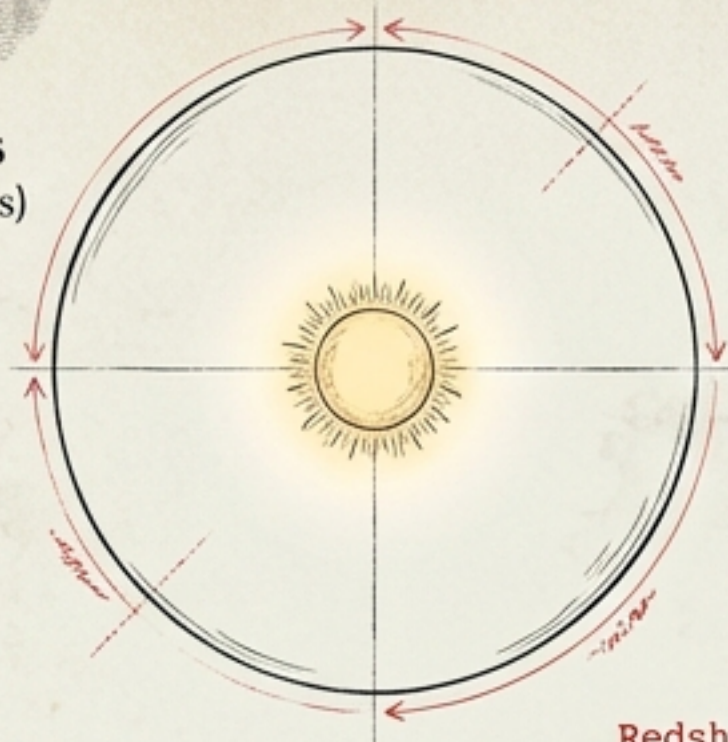
Ptolemy elaborated this into a cosmological model where the earth stood at the center, surrounded by eight spheres carrying the moon, sun, and planets.

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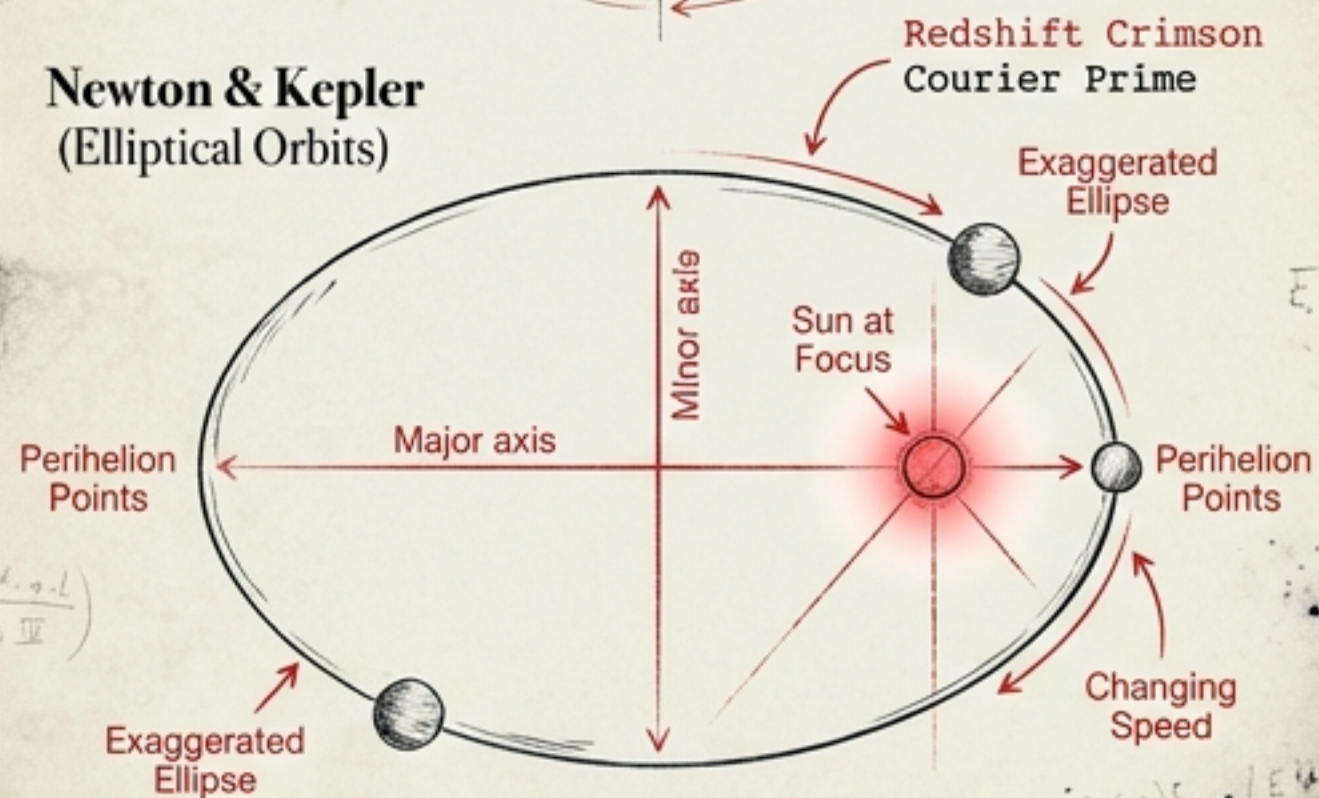
THE FLAW

To predict planetary positions correctly, Ptolemy had to assume the moon sometimes came twice as close to earth as other times—meaning it should appear twice as big. He ignored this flaw to preserve the model.

Copernicus
(Perfect Circles)



Newton & Kepler
(Elliptical Orbits)



Gravity and the Ellipse

Redshift Crimson

The death of the static center began in 1514 with Copernicus (Sun-centered) and was confirmed by Galileo and Kepler. But it was Isaac Newton's *Principia Mathematica* (1687) that provided the why.

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THE BREAKTHROUGH:

Redshift Crimson, Helvetica Neue

Newton proposed a law of universal gravitation: every body attracts every other body. The moon doesn't orbit because of a crystal sphere; it "falls" toward earth but misses it due to forward velocity.

NOTE:

Redshift Crimson

The story that Newton was inspired by an apple hitting his head is almost certainly apocryphal. All Newton himself ever said was that the idea came to him as he sat "in a contemplative mood" and "was occasioned by the fall of an apple."

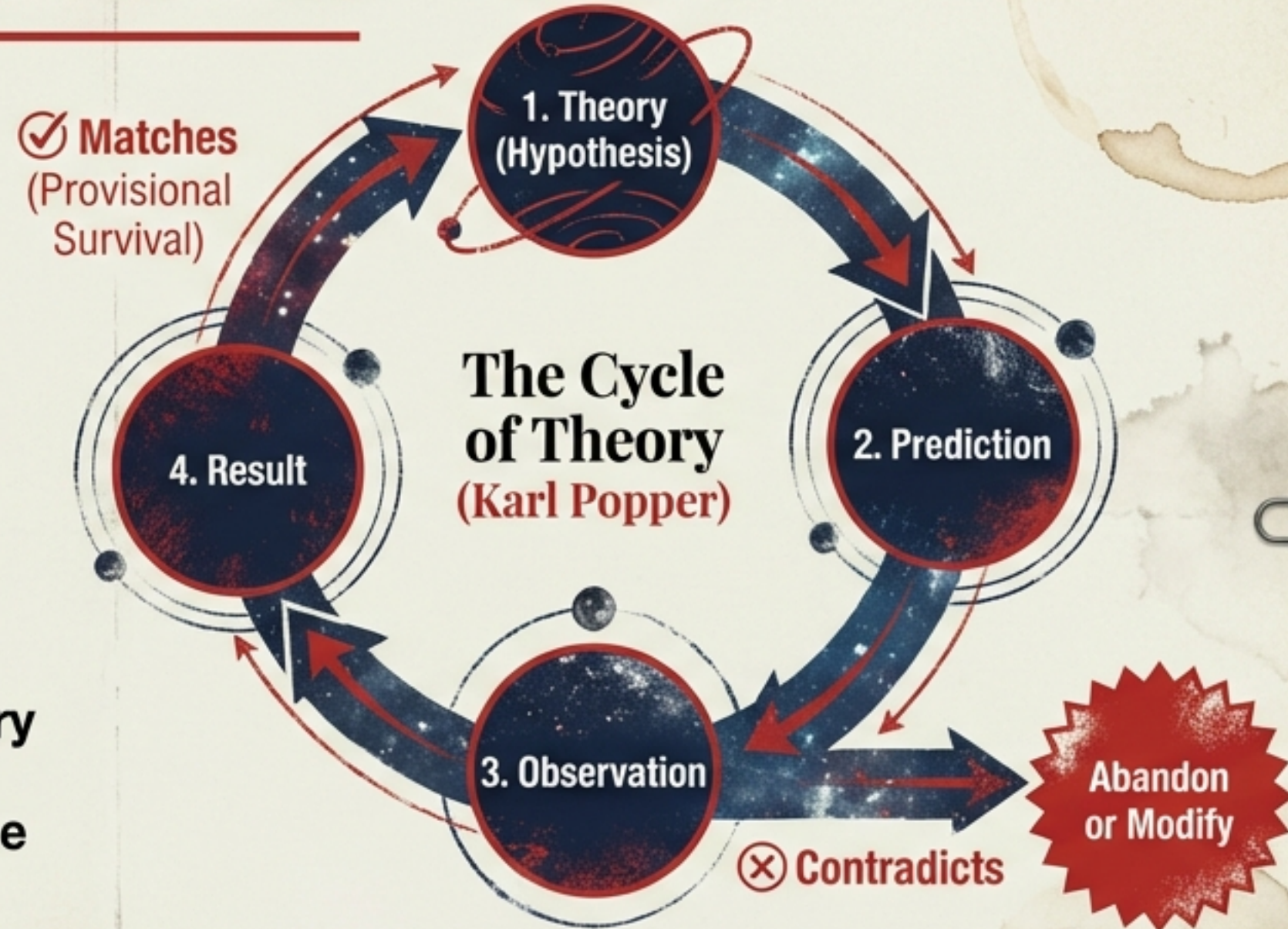
The Rules of the Game

In our search for reality, we must define what a "theory" actually is. It exists only in our minds.

A good theory must:

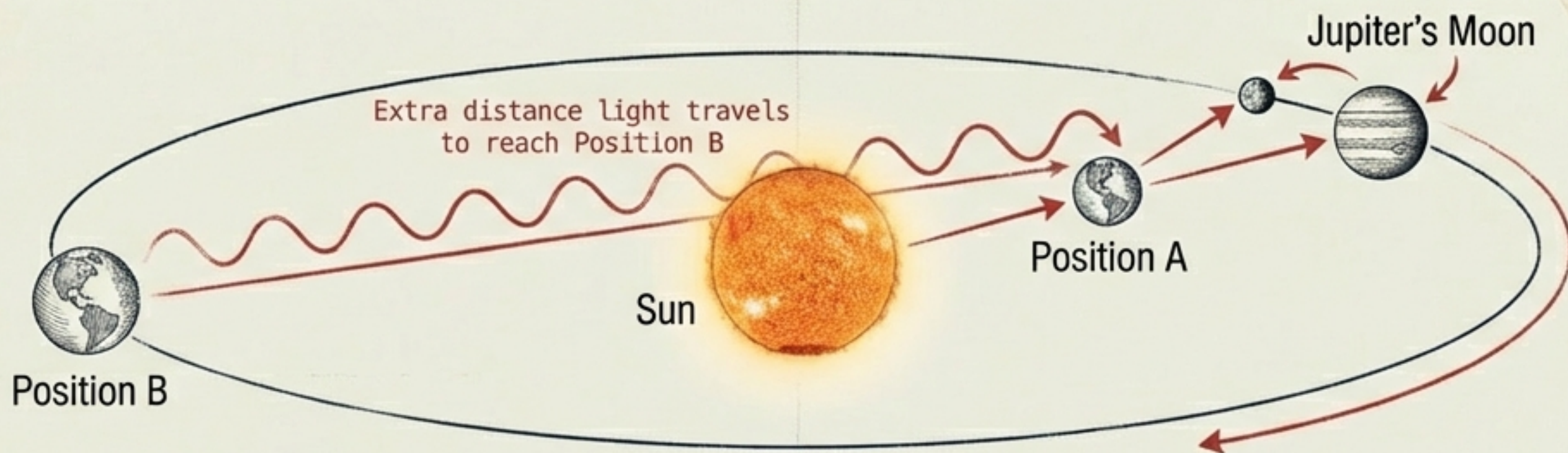
1. Accurately describe a large class of observations with few arbitrary elements.
2. Make definite predictions about future observations.

THE CATCH: You can never prove a theory (no matter how many times experiments agree), but you can disprove it with a single observation.



“Any physical theory is always provisional, in the sense that it is only a hypothesis: you can never prove it.”

The Finite Messenger



In 1676, Ole Christensen Roemer noticed that eclipses of Jupiter's moons appeared later when the Earth was farther away moon was farther away from Jupiter.

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THE CONCLUSION: Light is not instantaneous. It travels at a finite, albeit very high, speed (186,000 miles per second).

NARRATIVE SHIFT: This discovery lay dormant until Maxwell's equations (1865) unified electricity and mental study magnetism, predicting waves that travel at a fixed speed. But fixed relative to what?

The Ether Wind That Never Blew

Scientists assumed light waves must travel through a medium called "ether," just as sound travels through air. We should be moving through this ether, creating an "ether wind."

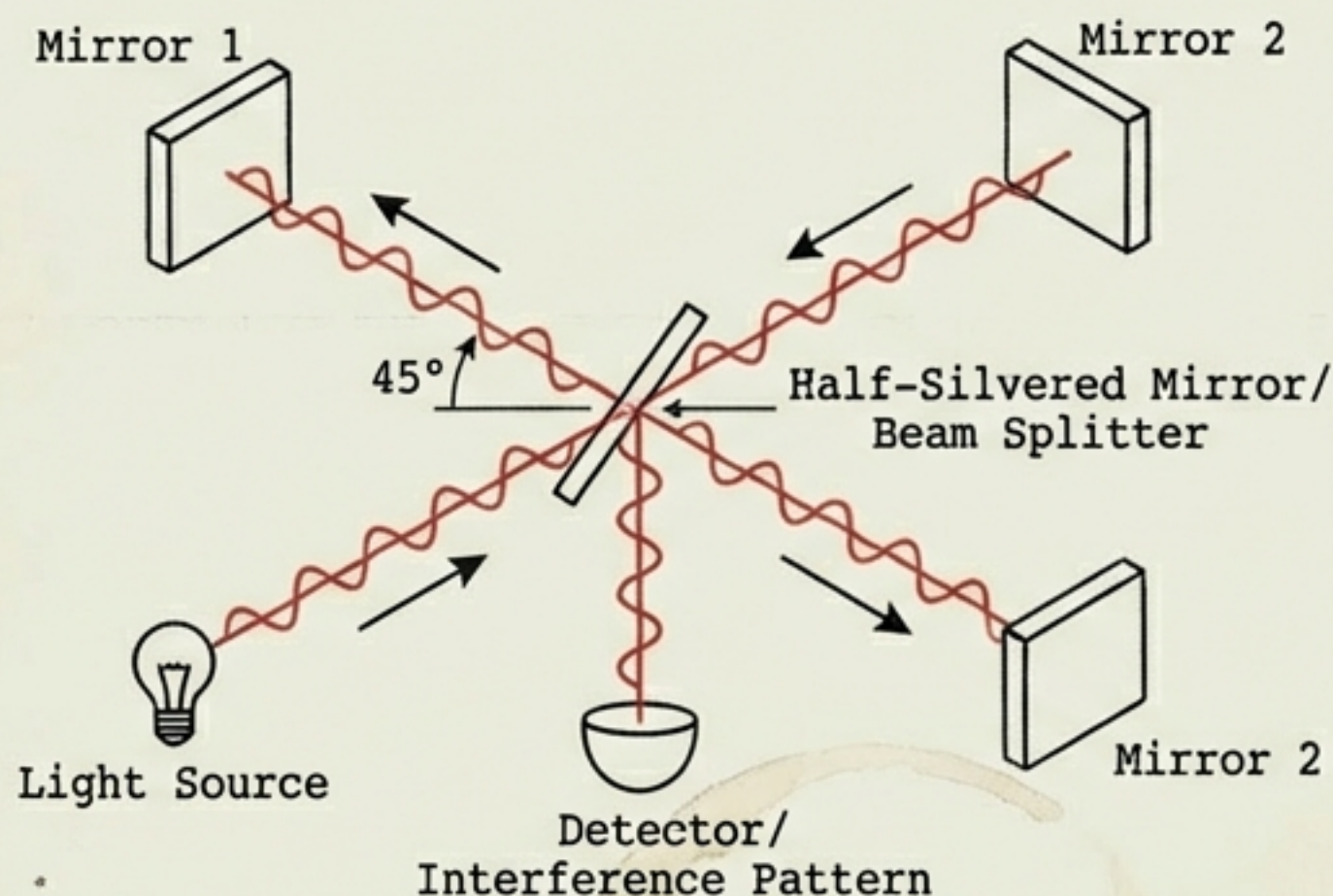


Fig. 3: Michelson-Morley Interferometer Schematic. The device was rotated to detect the "Ether Wind".

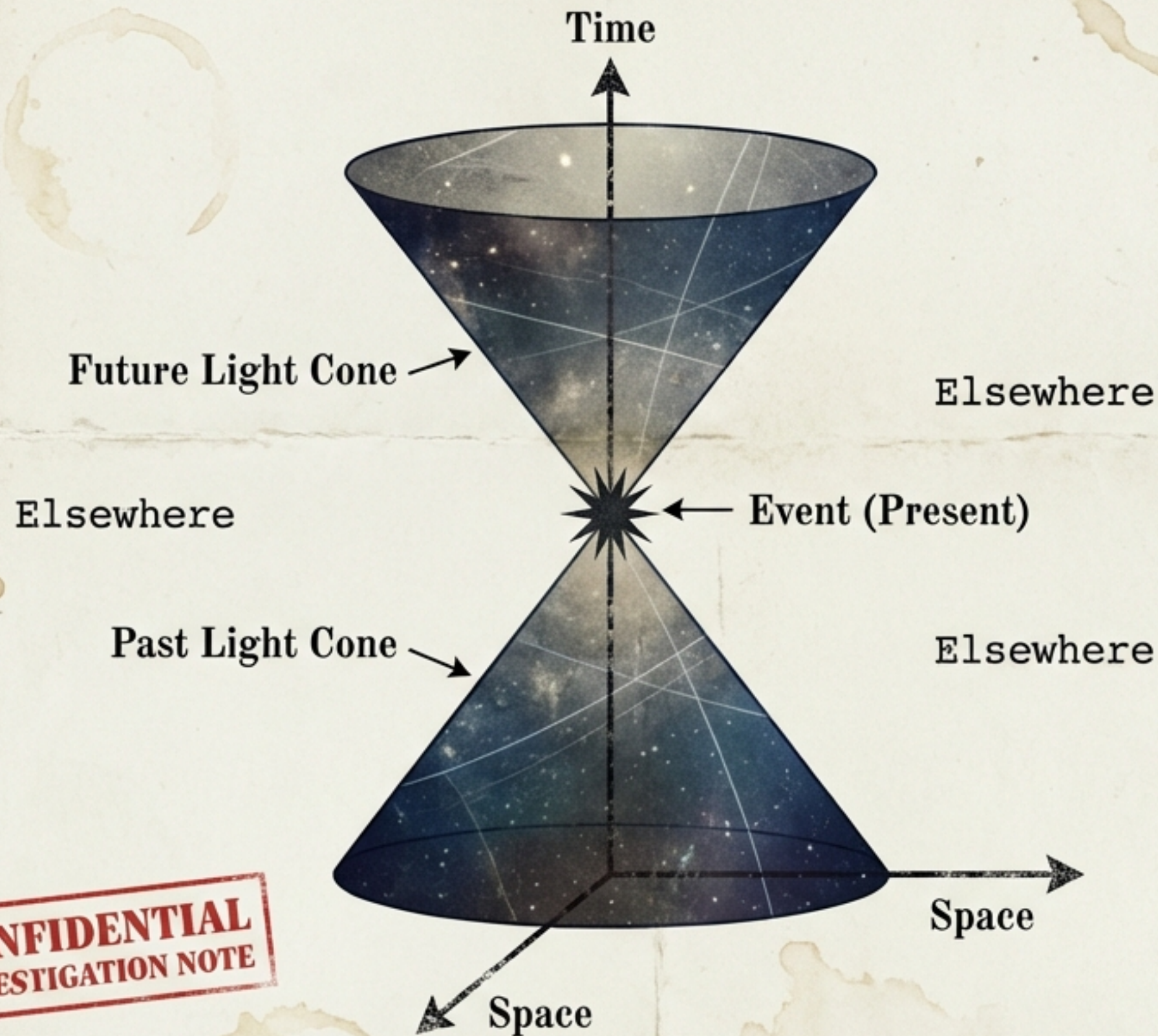
THE EXPERIMENT: In 1887, Michelson and Morley tried to measure the speed of light in the direction of the earth's motion vs. at right angles to it.

THE RESULT: The speeds were exactly the same. There was no ether.

THE IMPLICATION: If the speed of light is constant for everyone, regardless of how fast they are moving, then time and distance must be the things that change.

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Past, Future, and Elsewhere.



We must abandon "Absolute Time."
Events are defined by their position
in Space-Time.

KEY CONCEPT: THE LIGHT CONE.

1. Absolute Past: Events that could have affected the present.
2. Absolute Future: Events the present can affect.
3. Elsewhere: Events that cannot affect or be affected by the present because light hasn't had time to travel between them.

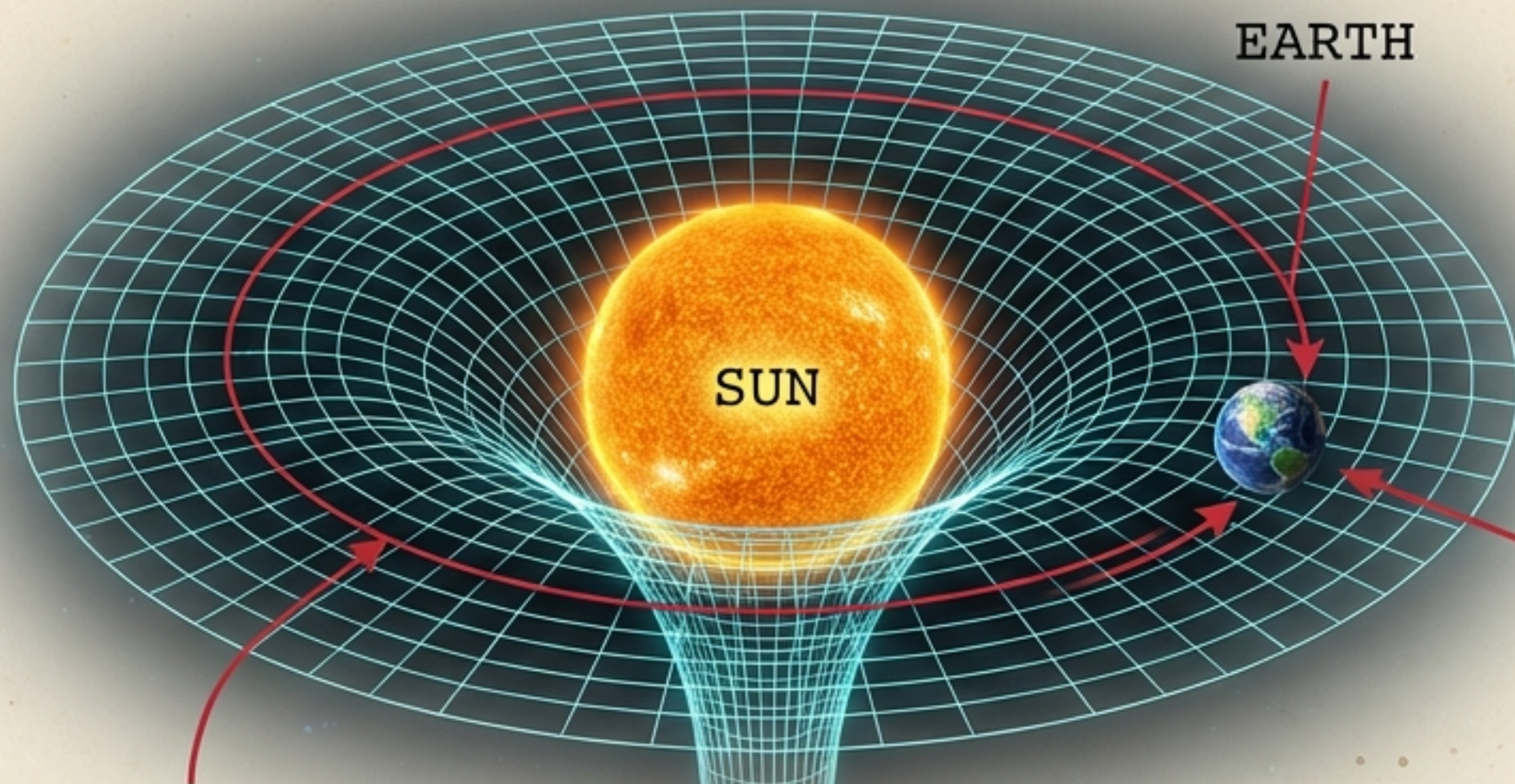
EXAMPLE:

If the sun went out right now, we wouldn't know for 8 minutes. The event is in our "elsewhere" until the light cone reaches us.

The Curvature of Space-Time

Einstein's General Theory of Relativity (1915) revolutionized gravity.

Gravity is not a force that attracts; it is the result of mass curving space-time.



PROOF:

The 1919 eclipse expedition proved that light from distant stars is bent by the sun's curvature.

PARADOX:

The "Twins Paradox"—a twin living on a mountain (further from earth's mass) ages faster than a twin at sea level.

THE GEODESIC:

Bodies like the earth try to move in straight lines through space-time, but because space-time is curved by the sun's mass, they appear to us to move in circles.

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The Fingerprint of Stars

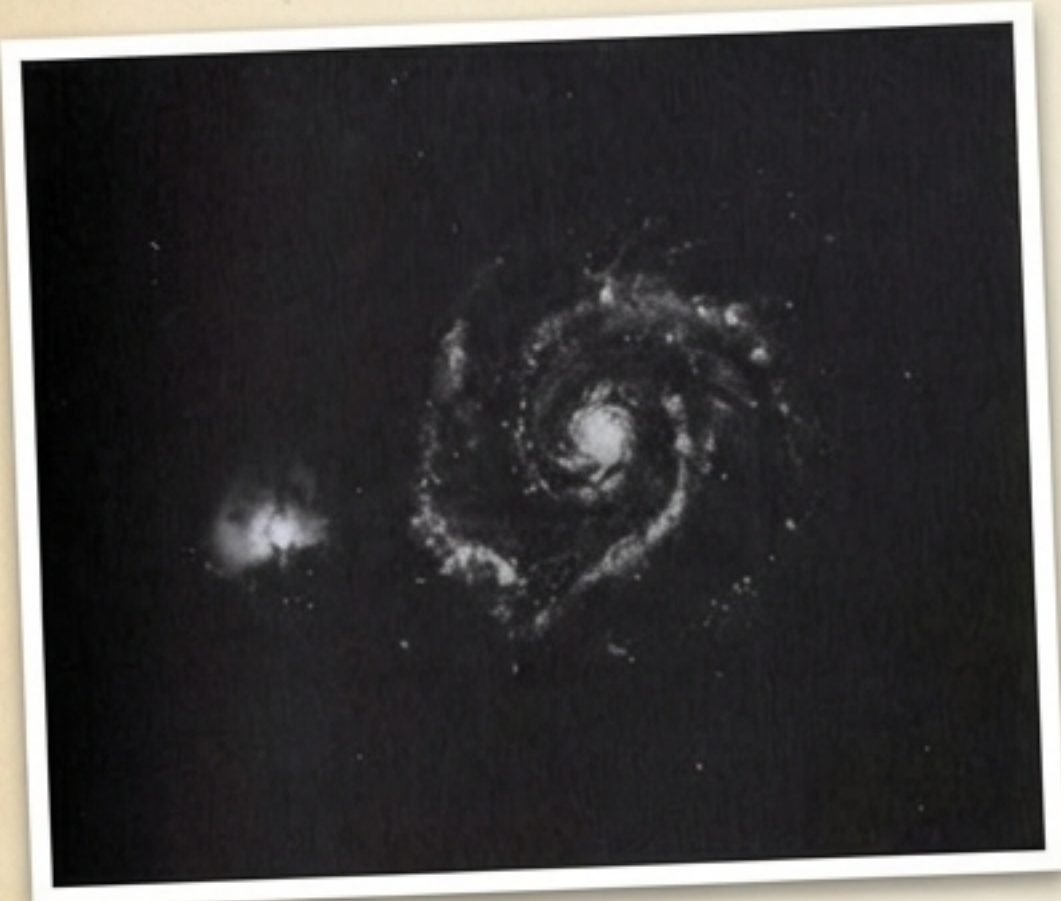


EXHIBIT A: SPIRAL GALAXY
(DISTANT SOURCE)

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THE "RED SHIFT" EVIDENCE



STANDARD SPECTRUM
(LOCAL SOURCE)



RED-SHIFTED
SPECTRUM
(DISTANT SOURCE)

SHIFT TO RED

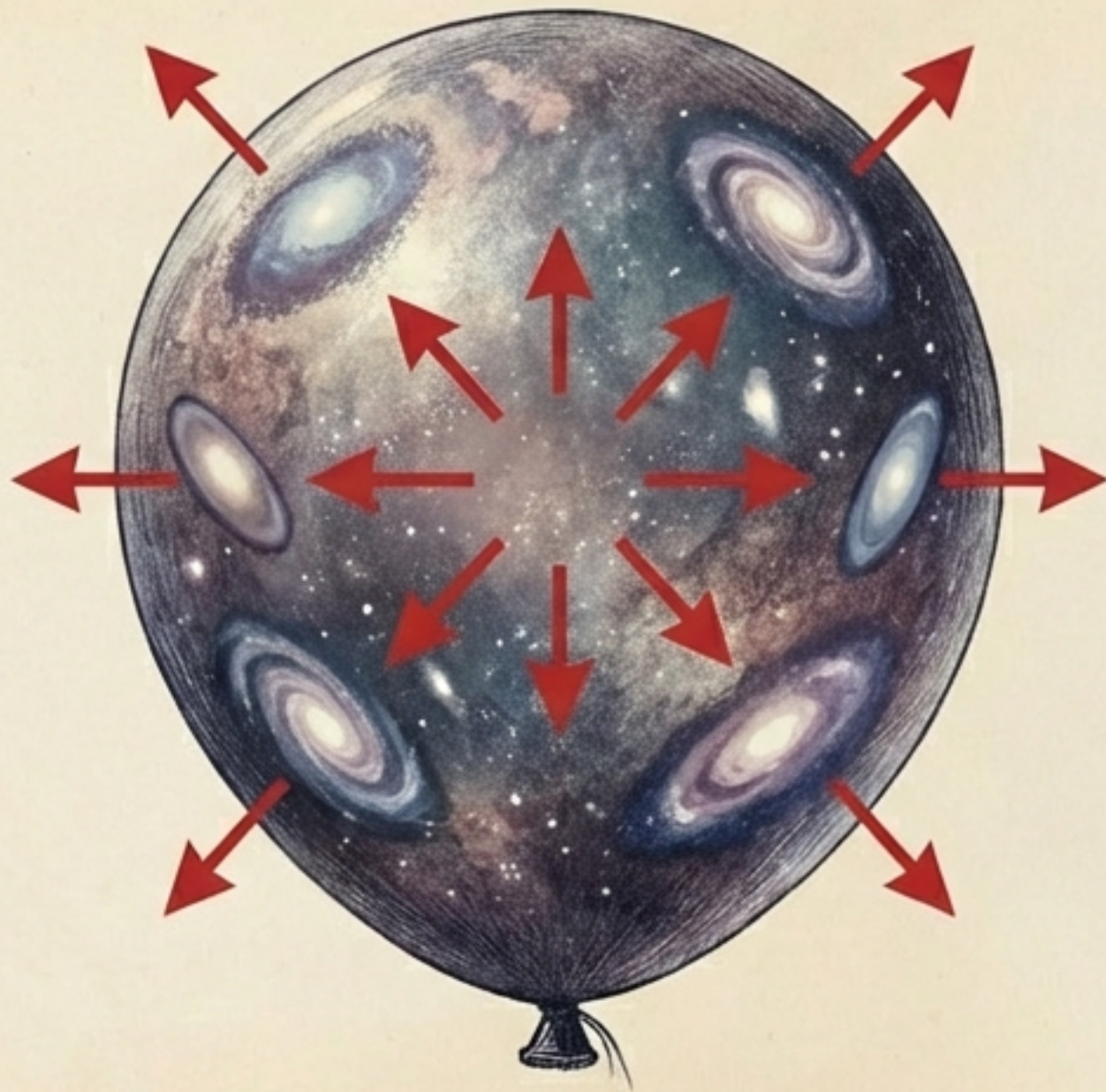
THE COSMIC FINGERPRINT: We can determine a star's constitution by the missing colors in its light spectrum. In the 1920s, astronomers noticed these missing colors were shifted toward the red end of the spectrum for distant galaxies.

THE DOPPLER EFFECT: Just as a siren sounds lower pitch as it moves away, light waves stretch (become redder) as a source moves away.

HUBBLE'S DISCOVERY (1929): The universe is expanding. The farther away a galaxy is, the faster it is moving away from us.

No Center, Only Expansion

Friedmann's Assumption:
The universe looks identical in whichever direction we look, and this would be true from anywhere else in the universe.



VISUAL EVIDENCE:
The Expanding Universe Model

THE ANALOGY

Imagine a balloon with spots. As you blow it up, every spot sees every other spot moving away. The farther the spots, the faster they separate.

IMPLICATION

There is no center to the universe's expansion. We are not special.

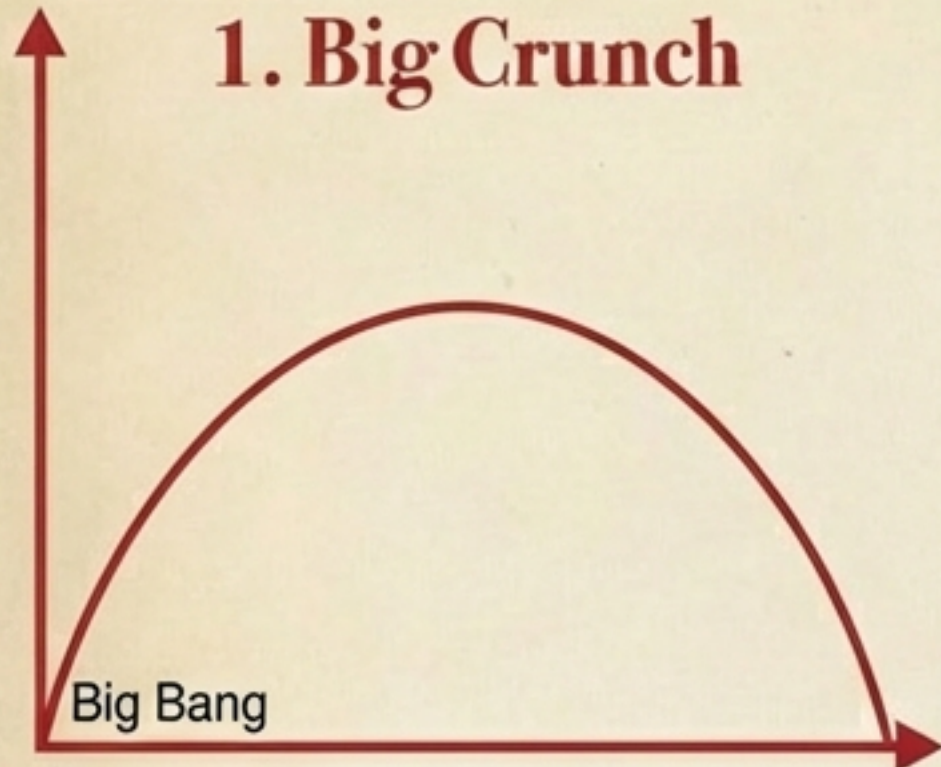
CONFIRMATION

The accidental discovery of Cosmic Microwave Background Radiation by Penzias and Wilson confirmed the universe is filled with the 'noise' of the early expansion.

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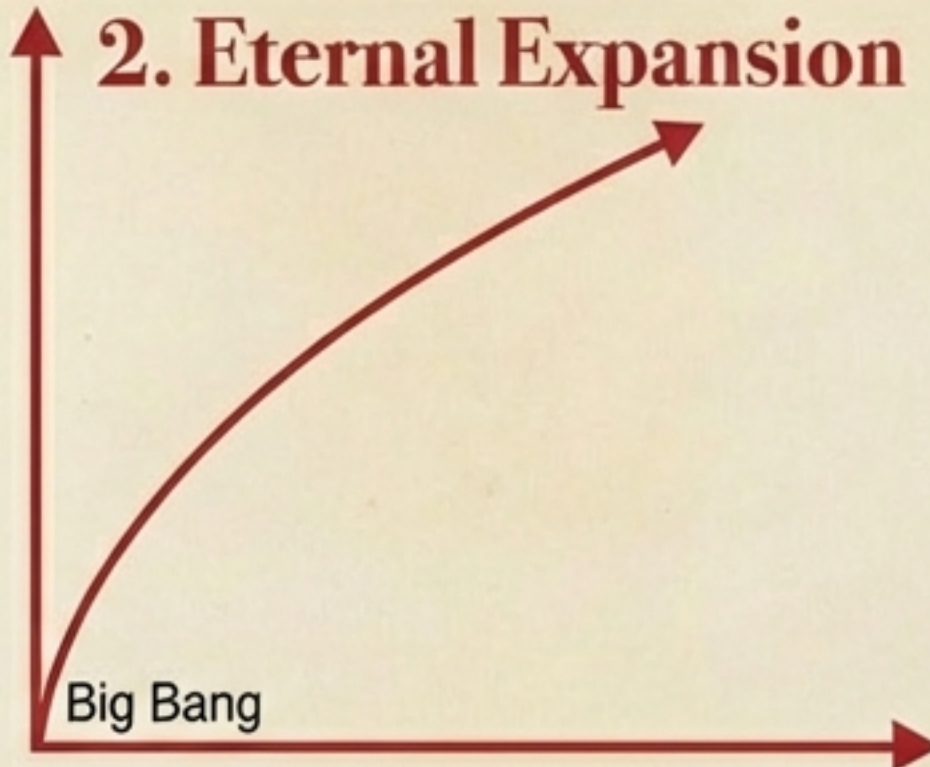
Three Fates

1. Big Crunch



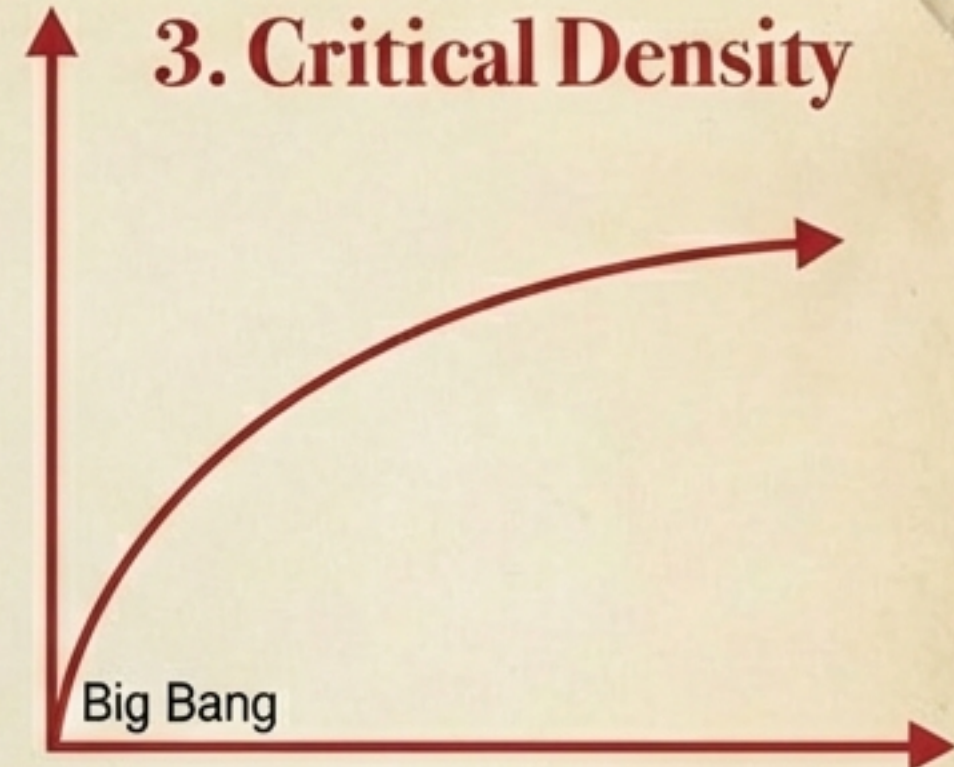
Gravity is strong enough
to stop expansion.
Space is finite/bent in.

2. Eternal Expansion



Gravity is too weak. Galaxies
drift apart forever. Space is
infinite/bent like a saddle.

3. Critical Density



The "Goldilocks" model.
Expansion slows but never
reaches zero. Space is flat.

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THE MISSING MASS: Current visible matter is only 10% of what is needed to stop expansion. We are searching for 'Dark Matter' to determine our fate.

The End of the Clockwork Universe



Laplace once dreamt of “Scientific Determinism”: If we knew the position and speed of every particle, we could predict the future perfectly.

THE FAILURE: This requires measuring position and speed precisely. But Planck's “Quanta” theory and Heisenberg's Uncertainty Principle destroyed this dream.

HEISENBERG'S LAW: To measure a particle, you must shine light on it. The light disturbs the particle. The more accurately you measure position, the less accurately you can measure speed.

“God plays dice.”

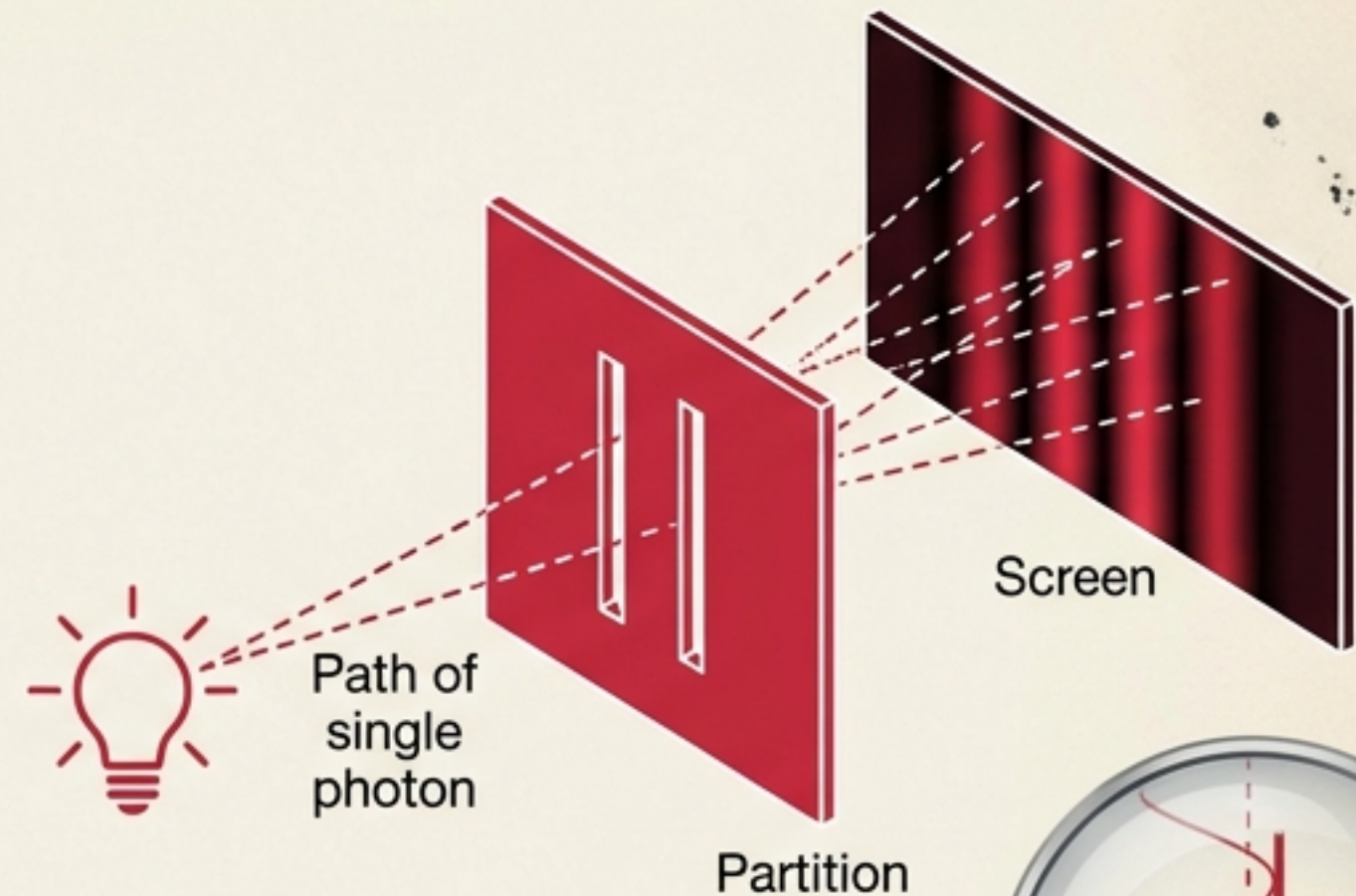
(Einstein's objection to this randomness)

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Waves, Particles, and Interference

Quantum mechanics introduces “Wave/Particle Duality.” Particles behave like waves, and waves behave like particles.

- **THE EXPERIMENT:** If you send electrons through two slits, they don't form two lines (like bullets). They form **interference fringes** (like water waves), canceling and reinforcing each other.
- **THE WEIRDNESS:** This happens even if you send electrons one at a time. The electron essentially passes through both slits at once, interfering with itself.
- **SUM OVER HISTORIES:** A particle doesn't have a single path; it takes every possible path simultaneously.



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THE OBSERVER EFFECT: Measuring which slit the particle goes through collapses the wave function, destroying the interference pattern. The act of observation changes the reality.

The Building Blocks

The Atom is not the smallest unit. Protons and Neutrons are made of Quarks (Up, Down, Strange, Charmed, Bottom, Top).



(i) **Spin = 0**

Looks same from every direction (a dot).

(ii) **Spin = 1**

Looks same after 1 turn (an arrow).

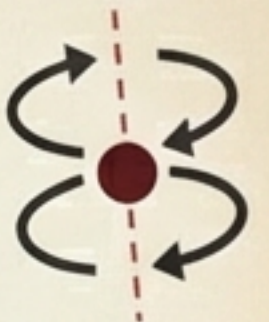


(iii) **Spin = 2**

Looks same after $\frac{1}{2}$ turn (double-headed arrow).

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SPIN $\frac{1}{2}$: The weird one. You must turn it two complete revolutions to look the same. These make up the matter of the universe.



The Four Forces

• 1. GRAVITY:	The weak giant. Long-range, always attractive. Carried by the 'Graviton' (Spin 2).
• 2. ELECTROMAGNETIC:	Holds atoms together. Attractive/Repulsive. Carried by the 'Photon' (Spin 1).
• 3. WEAK NUCLEAR:	Causes radioactivity. Short range.
• 4. STRONG NUCLEAR:	The glue of quarks. Holds the nucleus together. Carried by 'Gluons'.

THE GOAL: Grand Unified Theories (GUTs) attempt to merge these (except gravity) into a single force that existed at the Big Bang.

The Death of a Star

A star is a balance between heat (expanding) and gravity (compressing). When fuel runs out, gravity wins.



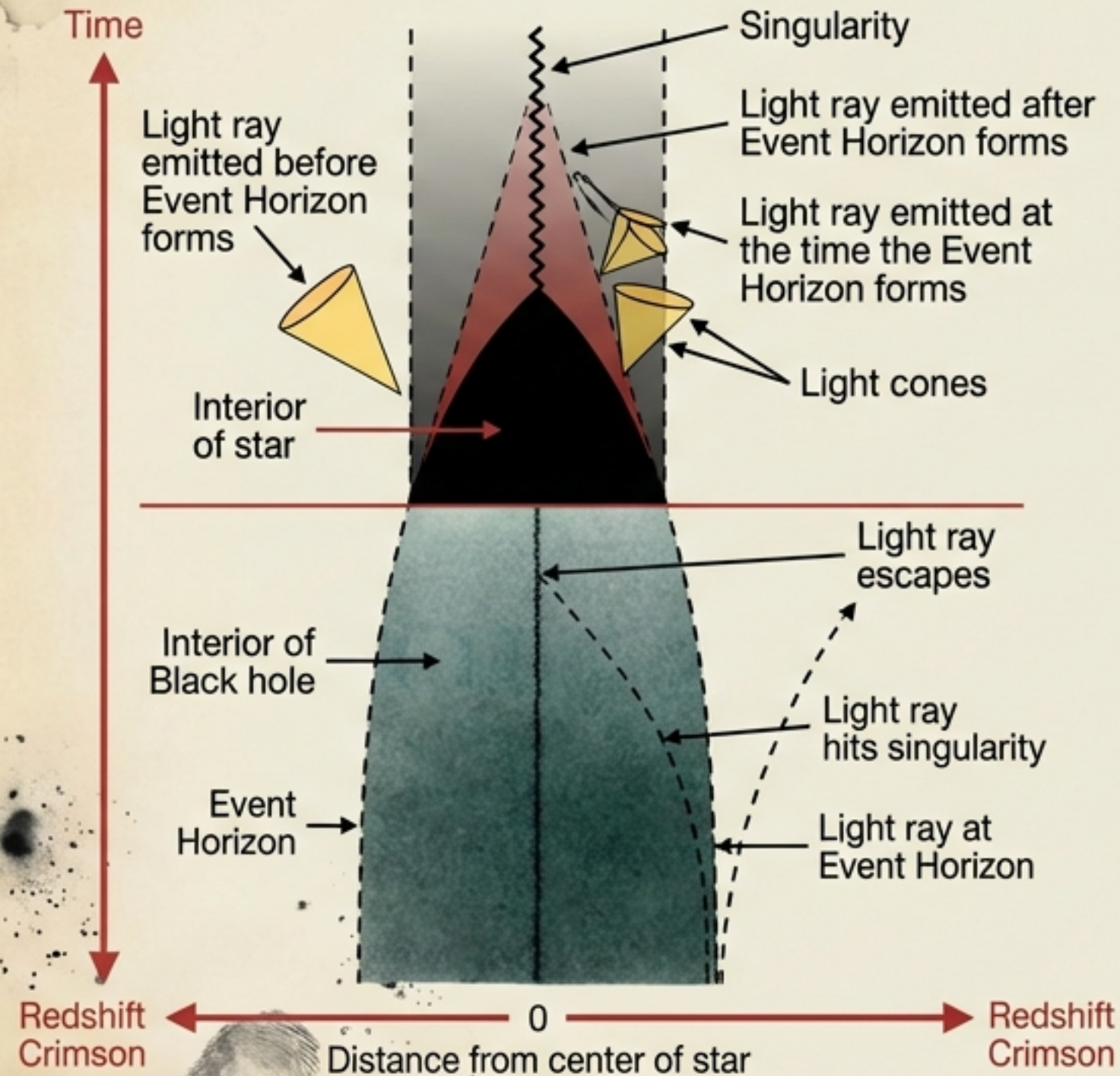
CHANDRASEKHAR LIMIT: If a cold star is more than 1.5x the mass of the sun, the repulsion of electrons (exclusion principle) cannot stop the collapse.

THE RESULT: The star collapses to zero size. Infinite density. A Singularity.

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"God abhors a naked singularity."
- Penrose's Cosmic Censorship Hypothesis

The Event Horizon

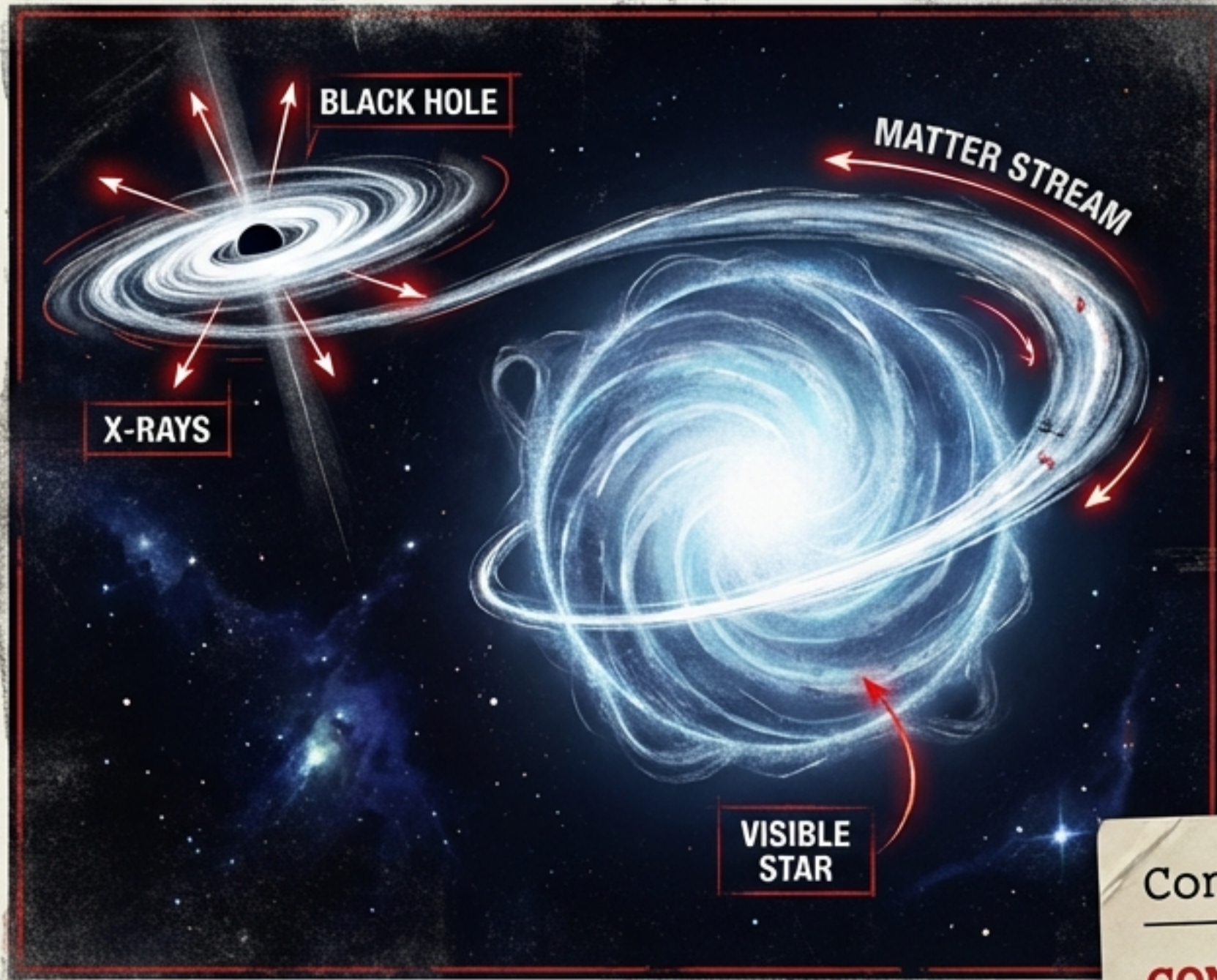


As a star collapses, its gravity bends space-time so severely that the “Light Cones” tip inward.

THE POINT OF NO RETURN: The Event Horizon is the boundary where light rays just hover, unable to escape but not falling in.

RELATIVE TIME: To an astronaut falling in, time passes normally. To a distant observer, the astronaut’s time slows to a halt, and their image fades to red and disappears.

Seeing the Invisible



We cannot see a black hole, but we can see its gravity torturing its neighbors.

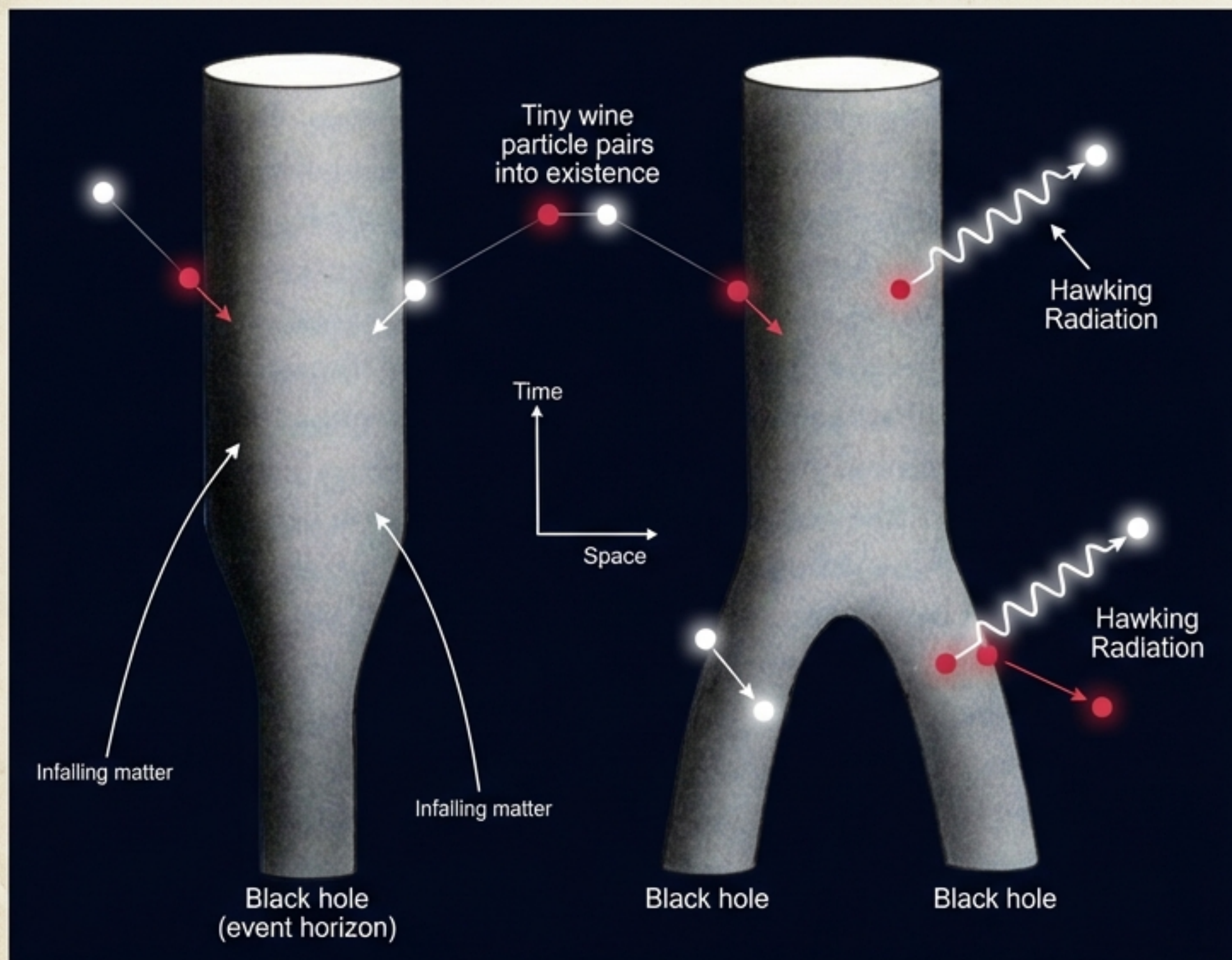
CASE STUDY: Cygnus X-1

- A visible star orbiting an unseen companion.
 1. The visible star is being whipped around by a massive object (6x the sun).
 2. Matter is being stripped off, spiraling into the void.
 3. As matter falls, it heats up and emits massive amounts of X-rays.

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CONCLUSION: It is too heavy to be a white dwarf or neutron star. It must be a black hole.

"Black Holes Ain't So Black"



THE PROBLEM: If a black hole absorbs order (entropy), does it violate the Second Law of Thermodynamics?

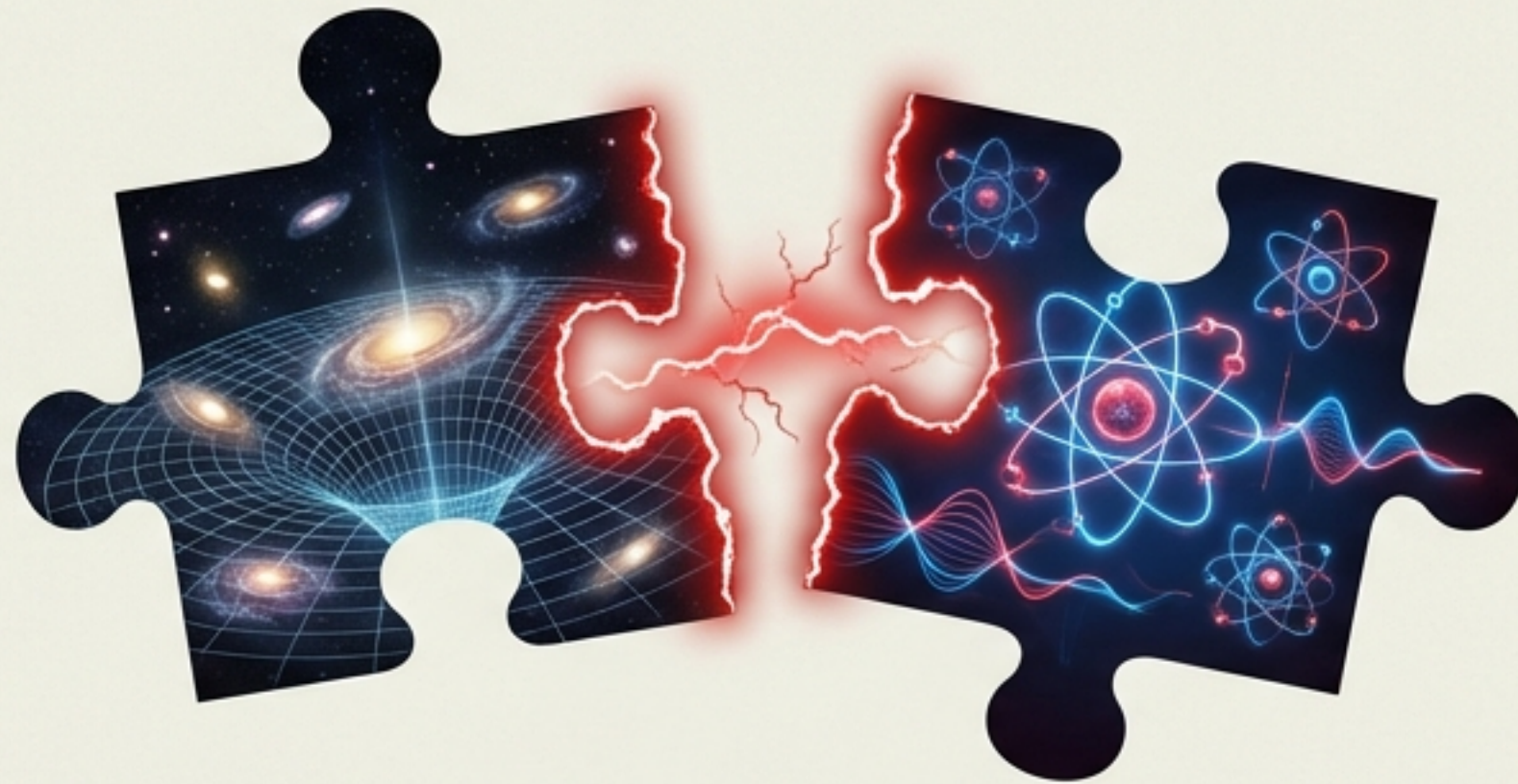
THE SOLUTION: The area of the event horizon represents entropy.

HAWKING RADIATION: Virtual particle pairs appear in space. Near the horizon, one falls in (negative energy) and one escapes (positive energy).

THE IMPLICATION: The escaping particle looks like radiation coming FROM the black hole. The black hole loses mass, glows, and eventually evaporates.

“The Quest for Unification”

**General
Relativity**
(The Big)



**Quantum
Mechanics**
(The Small)

We have two partial theories: **Relativity** and **Quantum Mechanics**.
They are inconsistent with each other.



THE CHALLENGE: They collide at the Singularity (Big Bang and Black Holes).
To understand our origin, we need a **Quantum** Theory of Gravity.

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“If we discover a complete theory... it would be the ultimate triumph
of human reason—for then we would know the mind of God.”